

A Machine Learning Model for the Prediction of Hessian Matrices based on Semiempirical Electronic Structure Theory

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Introduction

- The Hessian matrix is a target of interest, due to its high computational cost and its versatile applications.
- Atomistic Features, based on GFN2-xTB quantities, are used for the prediction of the target: Hessian matrix elements
- The features are permutation and rotational invariant to achieve a unique mapping onto the target.
- The zero point vibrational energy is used to quantify the precision of the model.

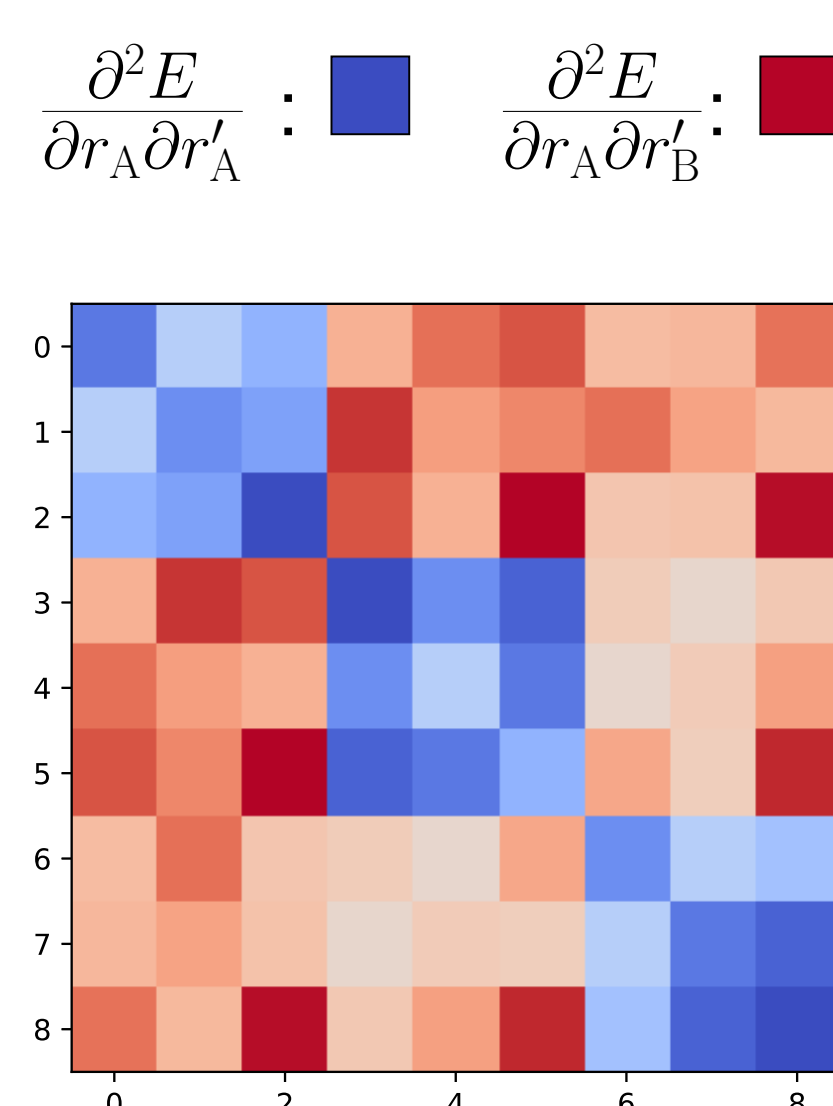
Overview of the ML Model

Origin of the Features

- Atomwise partitioned GFN2-xTB quantities are used for the features and were developed in our group
- The features are physics based and can be categorized into:
 1. Geometry: e.g. Coordination Number
 2. Density: e.g. Dipole Moment
 3. Energy: e.g. Coulomb-repulsion energy

The Hessian Matrix

- A $3N \times 3N$ matrix of second derivatives of the energy in respect to the coordinates of the N atoms
- The derivatives are either in respect of the coordinates of a single atom or of two atoms
- Two submodels are used to describe the diagonal blocks (blue) and off-diagonal blocks (red)

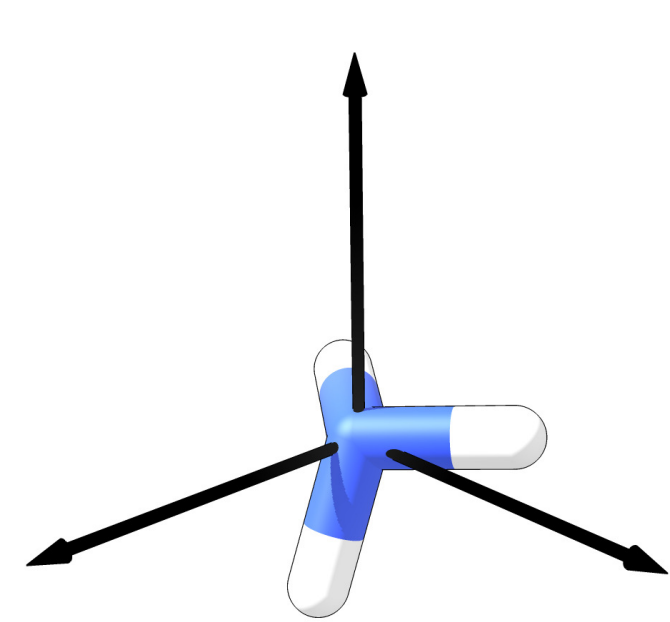
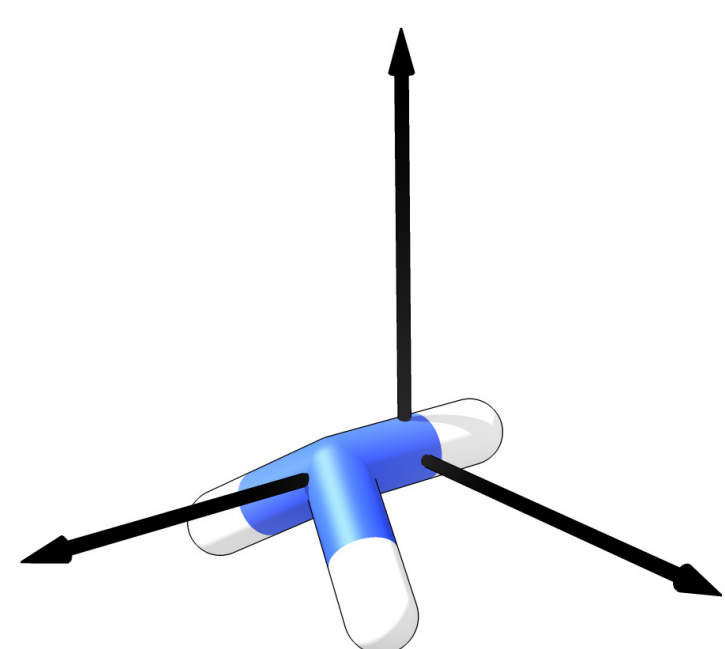


Connecting Features and Target

- The feature vector is either build up from the quantities of one or two atoms
- For ML modeling a constant feature vector length is necessary, hence two submodels are used
- The features and targets are computed for each 3×3 block in a different orientation.
- The molecular structure is oriented differently in the coordinate system in dependence of the submodel

Model  : Here the origin is shifted to the center of two atoms and the atoms are aligned to the z-axis.

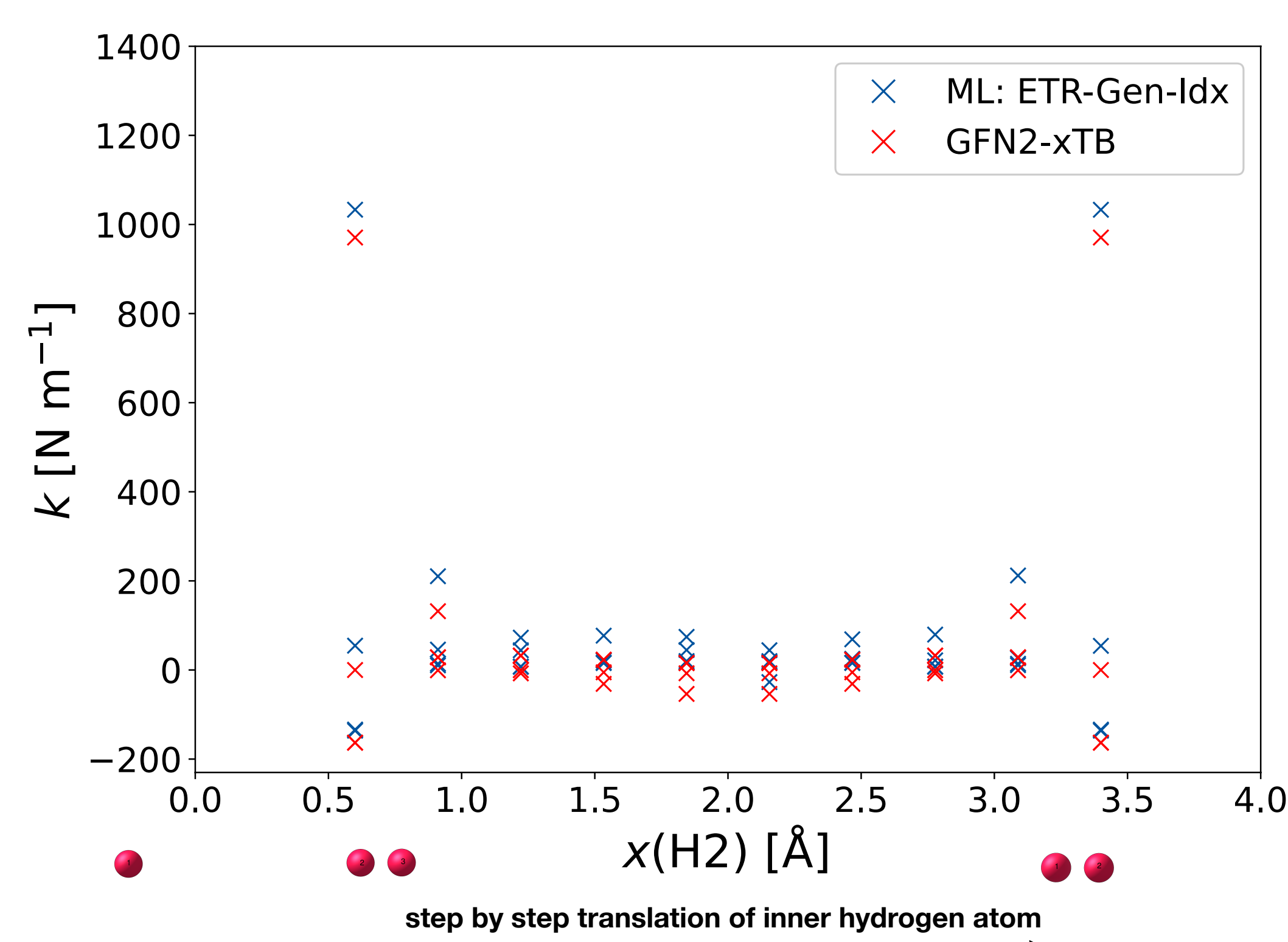
Model  : The origin of the coordinate system is shifted to the atom.



Unifying the Mapping of the Features

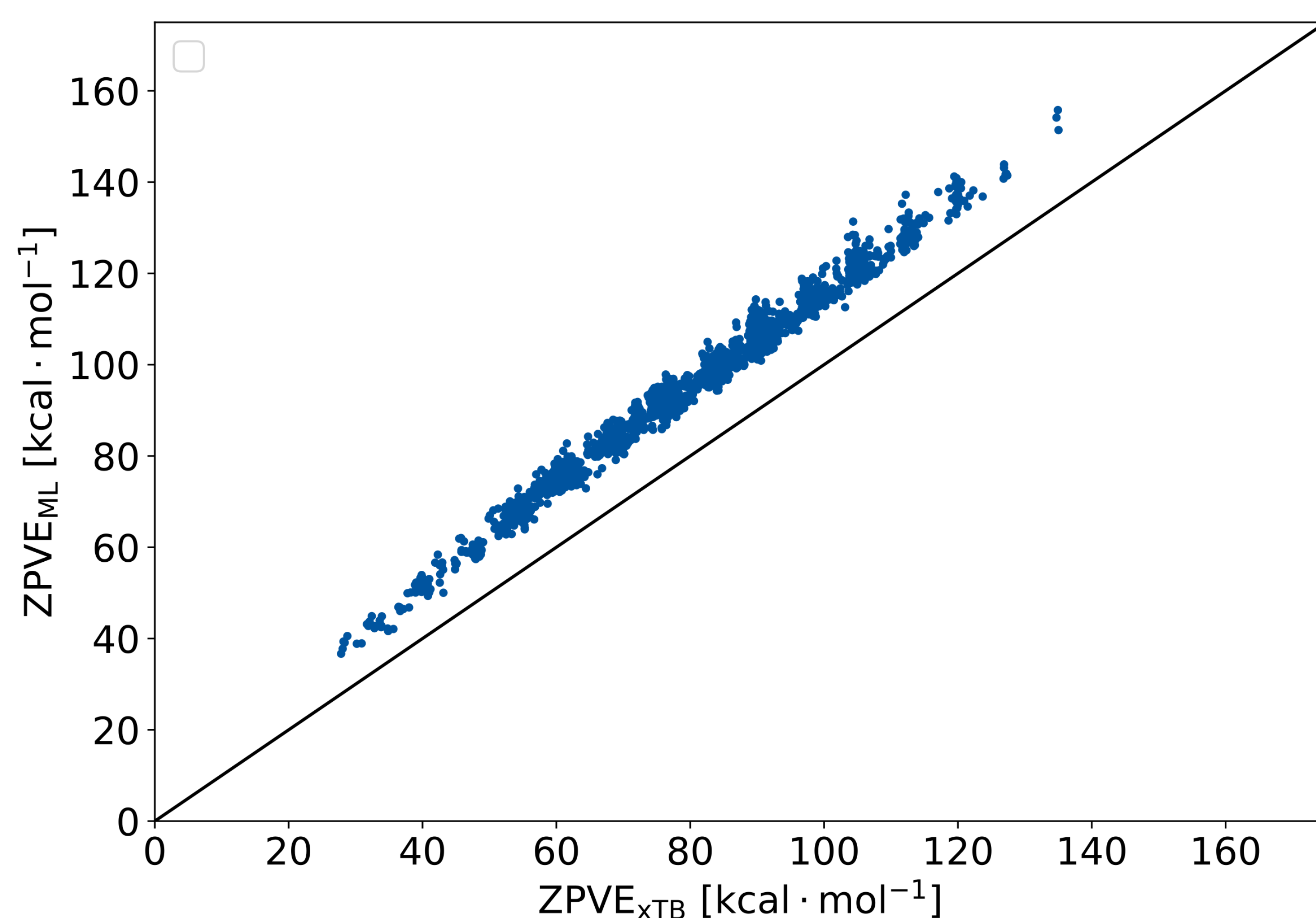
- A unique mapping of the features onto the target is achieved by different means
- An index is introduced to differ between the derivatives in respect to different coordinates of the same atom
- It is taken care of a consistent alignment to the atom oriented coordinate system

Verification of Unique Mapping



Predicted ZPVE of GBD7 Set

- Learning plots were computed using the GBD7 set, which consists of 7164 structures
- The test set was fixed to 25 % of the size of the GBD7 set
- The training set was increased up to 15% of the



Conclusion and Future Perspective

- Bullet point A
- Bullet point B
- Bullet point 3